



TBot-SA₁: a 3D-Causal World-Spatial-Action Model for Generalizable Robot Control

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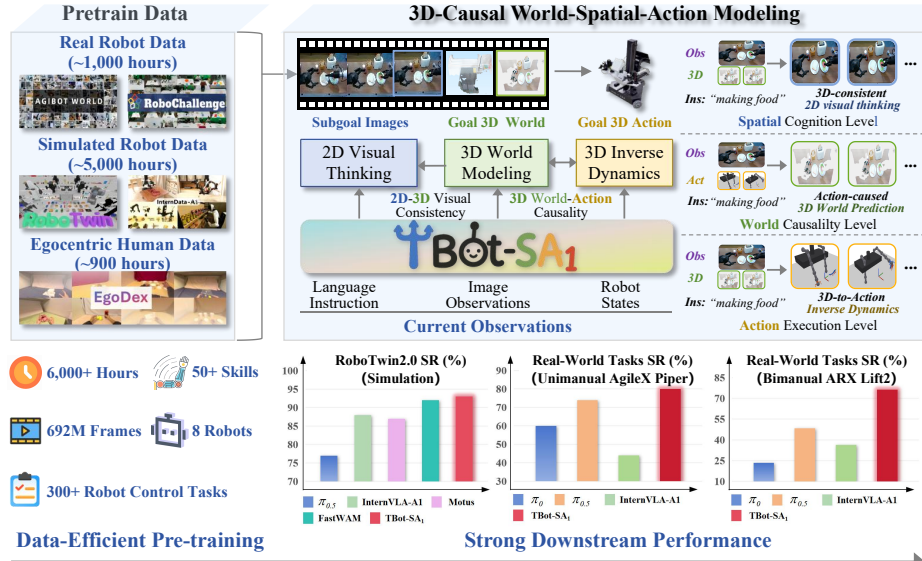


Figure 1: **TBot-SA₁**: A generalizable robot foundation model built upon 3D-Causal world-spatial-action modeling, achieving highly competitive manipulation performance across diverse simulated and real-robot benchmarks using only 6K hours of pre-training data.

Abstract

Recent advances in embodied AI have established robot foundation models (RFMs) as the dominant approach for generalist robotic systems to date. By leveraging imitation learning on extensive robot demonstrations, RFMs have achieved impressive capabilities in mapping visual observations and language instructions to continuous robotic actions. However, current RFMs lack an inherent ability to reason about physical dynamics and the causal effects of robot behaviors on the 3D physical world. This creates a fundamental mismatch between 2D-centric visual perception and 3D-centric embodied interaction, severely limiting the generalization ability of RFMs in real-world tasks. To address this gap, we present **TBot-SA₁**, a novel RFM built upon proposed 3D-Causal World-Spatial-Action (WSA) modeling paradigm. It not only learns 3D world-aware visual thought for future robot behaviors, but also models bidirectional causal relationships between 3D world state transitions and robotic actions to enhance behavior generalization. Notably, TBot-SA₁ achieves highly data-efficient pre-training with 6k hours of expert demonstration data (only 1k hours from real robot), while delivering competitive manipulation performance (93% success rate) on RoboTwin2.0 simulation benchmark and achieving +20% average boosted performance over state-of-the-art RFMs on real-world robot control tasks. These results reveal that generalizable RFM can be attained without large-scale real robot data when paired with 3D-centric causal world modeling, which offers a practical and affordable pathway to generalist robotic systems.

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1 Introduction

Creating versatile robots to match or exceed human-level physical intelligence has long been a core pursuit in embodied AI. At the core of human physical intelligence lies our innate ability to construct and maintain a spatial model of the 3D world. It allows us to parse the 3D structure of the world, reason about how our actions causally affect the state of the world, and execute dexterous motor actions to interact with the world [12, 14]. This 3D-centric capability of world-spatial-action co-modeling is equally vital for generalist robots, given that all physical interactions in the real world inherently unfold in 3D space and obey fundamental physical laws. Thus, developing a robot foundation model that can master intrinsic causality between 3D world evolution and robot behaviors, is the critical breakthrough needed to unlock the full potential of embodied AI.

In recent years, multi-modal large models [11, 3, 42, 33] have achieved remarkable progress and, meanwhile, demonstrated a promising technical roadmap toward beyond human-level intelligence through training on web-scale data. Following this direction, modern robot foundation models have gradually evolved into two typical paradigms: Vision-Language-Action (VLA) models [21, 47, 34, 29, 7, 6, 16, 45] and World Action Models (WAM) [4, 25, 9, 17, 41, 38, 15]. VLA models construct semantic-centric robotic policies by augmenting pre-trained vision-language models (VLMs) with a lightweight action expert. This design allows VLAs to inherit powerful open-world visual understanding capabilities from pre-trained VLMs, while the appended action expert generates robot actions conditioned on semantic-level visual comprehension, following an "understand-then-execute" paradigm. In contrast, WAM learns imagination-centric robot policies based on a predictive video generation model that acts as a vision planner. Based on this "imagine-then-execute" paradigm, its action expert functions as an inverse dynamics model, which generates robotic control signals corresponding to the planned visual frames.

Despite the impressive progress of current VLAs and WAMs, two fundamental challenges still impede their generalization to real-world scenarios. **First**, from a model paradigm perspective, both VLAs and WAMs lack an explicit mechanism for modeling the 3D world dynamics and the bidirectional causality between robot actions and world changes. The absence of explicit 3D causal modeling renders them unreliable in real-world tasks that require 3D spatial cognition and robust adaptation to environmental changes. **Second**, in terms of model training, the major optimization strategy for robot foundation models is the imitation learning, which heavily depends on large-scale real-world expert demonstration data. Acquiring such data at scale is prohibitively expensive and labor-intensive [5]. This data scarcity further prevents models from learning the generalizable physical commonsense behind demonstrations, which is essential for developing generalist robots that can adapt to unseen scenarios [5, 4]. These challenges motivate us to study a significant question:

Is it possible to move beyond the prevailing VLA/WAM paradigms and build a fundamentally new 3D-centric robot foundation model that learns world-action causality from demonstration rather than merely replicating behaviors through imitation?

In this work, we present **TBot-SA₁**, a novel robot foundation model built on 3D-Causal World-Spatial-Action (WSA) paradigm. As illustrated in Figure 3, the WSA breaks the inherent limitations of prevailing paradigms via unification of three learning tasks. First, **at the spatial cognition level**, it enables 3D-consistent 2D visual thinking, which empowers a robot to dynamically adjust visual plan as the 3D world changes. Second, **at the world causality level**, it explicitly models action-caused 3D world prediction, endowing a robot with the ability to capture the causal consequence of robot behaviors on 3D world. Third, **at the action execution level**, it learns a 3D inverse dynamics model that maps the planned 3D world state transitions to executable robot actions, fully closing the loop between 3D spatial world modeling and real-world interaction. This integrated closed-loop design allows TBot-SA₁ to not only generate goal-directed actions, but also predict causal consequences of its actions on 3D world and dynamically adjust its behavior accordingly. In summary, our main contributions are three-fold:

- **Robot Learning Paradigm:** We propose 3D-Causal World-Spatial-Action (WSA) modeling, a novel robot learning paradigm that unifies 3D-consistent 2D visual thinking, causal 3D world modeling, and 3D inverse dynamics into a single model, addressing the limited generalization of prevailing VLA and WAM paradigms.

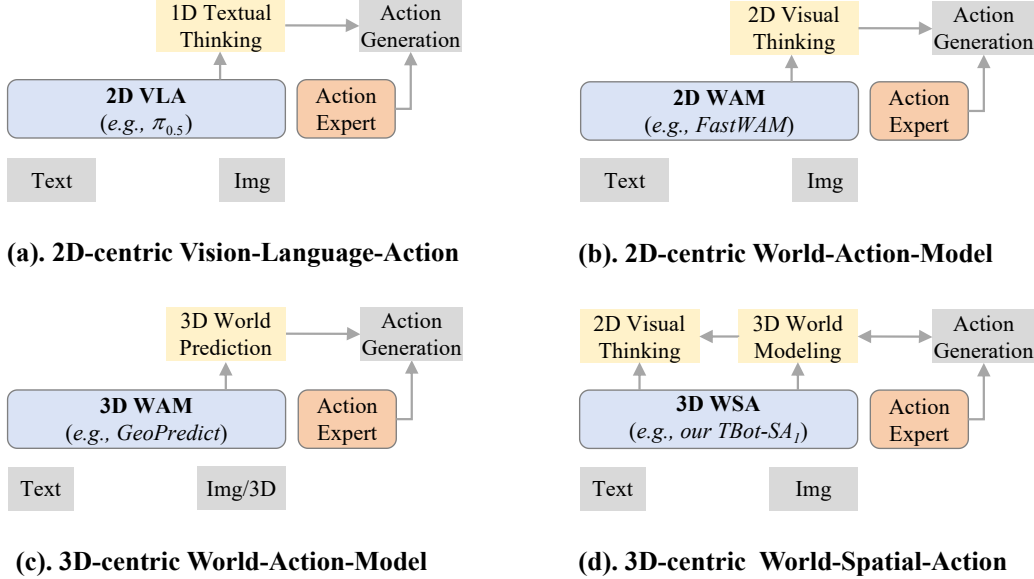


Figure 2: **Embodied Foundation Model Paradigms:** (a) 2D-centric VLA models learn unidirectional mappings from visual semantics to robot action; (b) 2D-centric World Action Model learns unidirectional causality from 2D visual dynamics to robot actions; (c) 3D-centric world action model learns unidirectional mappings from predicted 3D scenes to robot action; (d) 3D-Causal World-Spatial-Action model unifies bidirectional causality between 3D world dynamics and robot action into a single world.

- **Robot Foundation Model:** We present TBot-SA₁, the first robot foundation model built upon the 3D-Causal World-Spatial-Action (WSA) paradigm, which enables explicit bidirectional causality between 3D world dynamics and robot actions within a unified shared latent space.
- **Data-efficient Learning and Strong Performance:** We pre-train TBot-SA₁ with 6k hours of demonstration data (only 1k hours from real robot), and it delivers competitive manipulation performance (e.g., 93% success rate on RoboTwin2.0) on simulation benchmark and achieves +20% boosted performance on real-world robot control tasks.

2 Related Work

The core goal of robot foundation models (RFMs) is to establish a robust, stable multi-modal mapping from robot observations and task instructions to executable robot actions [43, 40]. As illustrated in Figure 2, mainstream research on RFMs has converged into two dominant paradigms: Vision-Language-Action models (VLAs) and World-Action-Models (WAMs). In the following sections, we provide a systematic literature review of these two research directions.

Vision-Language-Action: As the most widely adopted paradigm for robot manipulation, VLA models [34, 7, 6, 36, 45] are generally built on pretrained vision-language models. By inheriting rich semantic priors from VLMs [11, 3], they successfully transfer open-world visual-semantic knowledge to the physical robot domain via end-to-end fine-tuning on robot demonstration data. Early representative works such as RT-2 [47] and OpenVLA [16] adopt an autoregressive strategy, which discretizes continuous robot actions and formulates policy learning as a sequence modeling task. To address the limitation of discrete tokenization in fine-grained continuous control, recent state-of-the-art works [7, 6, 5] have shifted to generative frameworks based on conditional flow matching, which directly model the continuous-valued action, significantly improving the precision and stability of robot control. In addition, several works [46, 27] augment VLA with 3D geometry priors by incorporating extra 3D information such as depth or 3D gaussian splatting, demonstrating the effectiveness of 3D representation in robot learning. However, mainstream VLA policies are essentially reactive frameworks that make decisions based only on current observations, lacking

predictive planning and 3D causal reasoning capabilities, which makes them unreliable in real-world manipulation tasks that require 3D dynamics modeling.

World-Action-Model: In a different line from VLAs, WAMs follow the chain-of-thought philosophy and formulate robot policies as an imagine-then-execute procedure. Guided by this principle, numerous works [25, 38, 15, 41, 22] start with video models, inheriting rich commonsense knowledge from large-scale video generation pre-training. They then use these video models as world dynamics simulators, enhancing the reliability of robot foundation models via co-training of future frame prediction and action generation. For example, Motus [4] demonstrates that explicitly modeling future frame prediction significantly improves the generalization of robot manipulation, while Fast-WAM [41] proves that improved manipulative generalization comes from the co-learning rather than explicit video prediction. Furthermore, recent efforts [26, 44, 18] have further extended WAM paradigm to 3D space by modeling future geometry prediction. For instance, GeoPredict [26] leverages predictive kinematics and 3D Gaussian geometry to improve the manipulation performance, and PointWorld [14] scales 3D world models for in-the-wild robotic manipulation. However, existing 3D WAMs still follow a unidirectional "predict-then-execute" pipeline, failing to unify 3D world dynamics modeling and inverse dynamics into a single latent space for bidirectional causal learning. They cannot explicitly model how robot actions affect 3D space evolution and, in turn, how 3D space changes should guide action generation, which is the core focus of our work.

Compared with prior studies, our work is most closely related to WAMs but takes a further step: instead of directly co-learning video generation and robot control, our method unifies action-caused 3D world modeling and 3D inverse dynamics into a single model. This mechanism enables our model to exploit the intrinsic causal consistency: it not only learns how 3D robot actions affect the evolution of the 3D world but also determines the appropriate action plan given variations in the 3D world.

3 Methodology

In this section, we first formalize the robot control problem in Section 3.1. We then introduce the overall architecture of TBot-SA₁ in Section 3.2, followed by detailed learning objectives of the 3D-Causal World-Spatial-Action (WSA) modeling mechanism in Section 3.3. Finally, we present the pre-train data recipe in Section 3.4.

3.1 Problem Definition and Notation

In the context of robot control, the goal of embodied foundation model is to learn a robot policy that predicts next robot actions from current observations. Formally, at each timestep t , the observations include task instruction termed l , visual observation denoted as v_t , and robot proprioception termed s_t . In general, the robot action is a H -step chunk termed $A_t := (a_{t+1}, \dots, a_{t+H})$, a short-term action trajectory with a sequence of end-effector pose or joints of a robot in next H timesteps. To train a robot policy π_θ parameterized by θ on diverse robot demonstrations $\mathcal{D}_{\text{dem}}$, it is required to maximize the standard likelihood objective:

$$\max_{\theta} \mathbb{E}_{(v_t, s_t, A_t, l) \sim \mathcal{D}_{\text{dem}}} \log \pi_\theta(A_t \mid v_t, s_t, l). \quad (1)$$

As illustrated in Figure 2, previous works formulate robot control problem as vision-language-action or world-action policy, which can be regarded as a joint distribution derived from the standard objective.

2D Vision-Language-Action Policy. The 2D-centric VLA aims to directly predict executable actions conditioned on the current 2D observation $\mathbf{O}_t := \{v_t, s_t, l\}$. It learns 2D semantic-action unidirectional causality, which is formulated as a posterior distribution:

$$\mathbf{A}_t \sim p(a_{t+1:t+H} \mid v_t, s_t, l). \quad (2)$$

2D World-Action-Model Policy. The 2D-centric WAM couples predictive visual generation with next action generation, which jointly predicts executable actions and N subgoal images $\mathbf{V}_t := (v_{t+1}, \dots, v_{t+N})$ from the current observation $\mathbf{O}_t$. It learns 2D world-action unidirectional causality, which can be formulated as a joint prediction distribution:

$$\mathbf{A}_t \sim p(a_{t+1:t+H}, v_{t+1:t+N} \mid v_t, s_t, l). \quad (3)$$

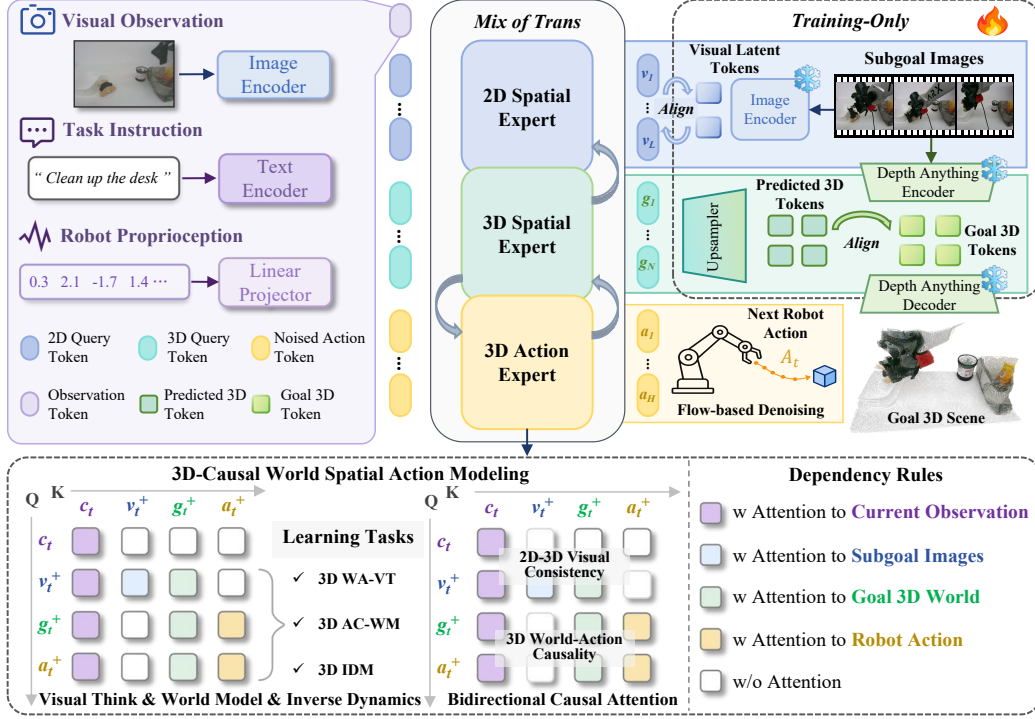


Figure 3: Overview of TBot-SA₁. It adopts a Mixture-of-Transformers structure with three complementary experts: a 2D Spatial Expert for 3D world-aware visual thinking, a 3D Spatial Expert for action-caused 3D world prediction, and a 3D Action Expert for 3D inverse dynamics modeling. A bidirectional causal attention mechanism enforces dependency rules to unify 3D-consistent visual thinking and bidirectional world-action causality within a shared latent space.

3D World-Action-Model Policy. Instead of modeling 2D world-action joint distribution, the 3D-centric WAM focuses on the unidirectional causality between evolution of scenes and robot action in 3D space. Specifically, given 2D observation $\mathbf{O}_t$ and 3D observation g_t , it learns to predict K subgoal 3D scenes $\mathbf{G}_t := (g_{t+1}, \dots, g_{t+K})$ for robot control. Formally, this models a joint prediction distribution as follows:

$$\mathbf{A}_t \sim p(a_{t+1:t+H}, g_{t+1:t+K} \mid g_t, v_t, s_t, l). \quad (4)$$

3D World Spatial Action Policy. Different from previous works, we formulate robot control as 3D-causal world-spatial-action learning, which jointly models 3D-consistent visual thinking and 3D world-action bidirectional causality. Given the current observations $\mathbf{O}_t$, the WSA models a joint distribution upon three prediction tasks that are relevant to robot control, including 3D goal-directed image prediction, action-caused 3D world prediction and 3D-conditioned action generation:

$$\mathbf{A}_t \sim p(\mathbf{V}_t, \mathbf{G}_t, \mathbf{A}_t \mid \mathbf{O}_t) = p(\mathbf{V}_t \mid \mathbf{O}_t, \mathbf{G}_t) \cdot p(\mathbf{G}_t \mid \mathbf{O}_t, \mathbf{A}_t) \cdot p(\mathbf{A}_t \mid \mathbf{O}_t, \mathbf{G}_t). \quad (5)$$

3.2 Model Architecture

Inspired by the recent works [4, 6], TBot-SA₁ is built upon a unified multi-modal transformer, which leverages mixture-of-transformer (MoT) architecture to enable seamless integration of 2D visual generation, future 3D prediction and robot action generation. As illustrated in Figure 3, it involves three major components: 2D spatial expert, 3D spatial expert and 3D action expert.

2D Spatial Expert (2D-SE): Given a human instruction, generating 3D goal-directed visual changes in 2D space enhances understanding of world dynamics. The 2D-SE $f_{2D}(\cdot)$ serves as the world-aware visual thinking branch of TBot-SA₁. It is initialized from a pre-trained vision-language model such as QWen3-VL [2] or Wan [30], thereby inheriting strong vision-language priors. The goal of 2D-SE is to predict future 2D visual dynamics to reach the target state of the 3D world.

Specifically, it predicts highly abstracted tokens $h_v = f_{2D}(O_t, G_t)$ for subgoal images, instead of RGB pixels. This design enhance visual understanding capability of the 2D-SE, as it is required to focus on essential features of an image rather than redundant pixel information.

3D Spatial Expert (3D-SE): Given a sequence of executable robot actions, predicting evolution of world dynamics in 3D space reflects causal effect of the robot behavior. Under this context, the 3D-SE $f_{3D}(\cdot)$ conducts a internal mechanism for 3D causal world modeling. To capture essential feature of 3D scenes, the 3D-SE predicts latent representations $h_g = f_{3D}(O_t, A_t)$ for subgoal 3D scenes. We construct it with a transformer-based module of the same depth as 2D-SE, thereby the architecture of 3D-SE can be directly derived from that of 2D-SE.

3D Action Expert (3D-AE): Given a demonstration of 3D space evolution, predicting next robot action tells why and how robot to behave in 3D space. The 3D-AE $f_{act}(\cdot)$ aims to predict a full action chunk conditioned on both current observations and future 3D latent representations. Specifically, we instantiate 3D-AE as denoising diffusion transformer. Given noisy actions $\hat{A}_t = \tau A_t + (1 - \tau)\omega$, where $\omega \sim \mathcal{N}(0, 1)$ and $\tau \in [0, 1]$, it employs iterative cross-attention and denoising procedure to predict latent action representations $h_{act} = f_{act}(O_t, G_t, \hat{A}_t)$. This yields a 3D inverse dynamics formulation, where 3D-AE maps predicted 3D latent evolution back to executable action.

3D-centric World-Action Causal Attention: Three experts are coupled within the mixture-of-transformer architecture via a shared attention mechanism. Formally, let h_t denote tokens of current observation, which are extracted via 2D-SE. As illustrated in Figure 3, we adopt a 3D-centric causal attention mechanism, which defines the following dependency rules: (1) All predicted tokens (h_v, h_g, h_{act}) can attend to the current observation tokens h_t , ensuring all predictions are grounded in the current state of the robot and the environment. (2) The visual tokens of future subgoal images h_v can attend to the 3D scene tokens h_g , encouraging the model to learn 2D-3D visual consistency and ground semantic visual predictions in physically consistent 3D geometry. (3) The 3D scene tokens h_g and action tokens h_{act} have full bidirectional attention access to each other, enabling joint learning of action-conditioned 3D dynamics prediction and 3D scene-aware action generation. This bidirectional attention is the key to bidirectional causal modeling of the WSA paradigm.

3.3 Learning Objective

TBot-SA₁ is optimized end-to-end with three joint training objectives, which correspond to the three factorized terms in the WSA joint distribution (Equ. 5) and align with the three core capabilities of the WSA paradigm:

3D World-Aware Visual Thinking (3D WA-VT). We adopt a MSE loss over the predicted visual tokens, to minimize the discrepancy between the predicted subgoal image tokens and the ground-truth tokens extracted via pre-trained tokenizer. Formally, the loss is defined as:

$$\mathcal{L}_{2D} = \mathbb{E}_{(O_t, G_t, V_t) \sim \mathcal{D}_{dem}} \|f_{2D}(O_t, G_t) - f_{enc}(V_t)\|_2^2, \quad (6)$$

where $f_{enc}(\cdot)$ denotes a pre-trained VAE-based tokenizer (e.g., COSMOS [1]).

Action-Conditioned 3D World Modeling (3D AC-WM). We adopt MSE loss over the predicted 3D scene latent representations, to minimize the error between the predicted 3D scene evolution and the ground-truth 3D geometric states from the demonstration data. The loss is formalized as:

$$\mathcal{L}_{3D} = \mathbb{E}_{(O_t, A_t, G_t) \sim \mathcal{D}_{dem}} \|f_{3D}(O_t, A_t) - f_g(G_t)\|_2^2, \quad (7)$$

where $f_g(\cdot)$ denotes a pre-trained 3D foundation model, e.g., Depth-Anything [19].

3D Inverse Dynamics Modeling (3D IDM). Following previous works [26, 6], we adopt flow matching loss [20] for continuous action generation. This loss minimizes the error between the predicted flow field and the ground-truth one, conditioned on the current observation and the 3D scene state evolution. The loss is defined as:

$$\mathcal{L}_{ACT} = \mathbb{E}_{(O_t, G_t, A_t) \sim \mathcal{D}_{dem}} \left\| \omega - A_t - f_{act}(O_t, G_t, \hat{A}_t) \right\|_2^2. \quad (8)$$

Table 1: Data mixture used for pretraining.

Data source	Type	Num. frames	Num. tasks	Sampling weight
InternData-A1 [31]	Sim.	396M	70	0.64
RoboTwin [10]	Sim.	17M	50	0.08
AgiBot-World [8]	Real	206M	217	0.18
RoboChallenge [35]	Real	5M	30	0.02
EgoDex [13]	Human	68M	194	0.08

This trains 3D-AE to predict flow field ($\omega - A_t$), thereby facilitating coherent action generation. The overall learning objective is the sum of the three losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{2\text{D}} + \mathcal{L}_{3\text{D}} + \mathcal{L}_{\text{ACT}}. \quad (9)$$

We adopt two-stage strategy to optimize the TBot-SA₁: pre-training on diverse data sources to acquire real-world behavior priors in first stage; post-training on specific manipulation tasks to adapt learned priors to target embodiment in second stage. Different from previous works [9, 4], TBot-SA₁ is trained by co-learning of three capabilities in both pre-training and post-training stage, rather than using a hierarchical manner.

3.4 Pre-training Data

High-quality embodied data is the cornerstone of pre-training generalizable embodied foundation models that can acquire versatile manipulation skills for flexible real-world robot control. However, due to the prohibitive cost and safety constraints of human-robot teleoperation, collecting large-scale high-quality real-world robot data at scale remains challenging [5]. To address this limitation, alternative embodied data sources such as simulation data and egocentric human data have emerged as critical complements. This gives rise to a multi-sourced data pyramid [5, 4], where data quantity decreases from the bottom to the top. It suggests that each data layer exhibits distinct, complementary strengths and inherent limitations, and no single data source alone can support the development of generalizable embodied foundation model: human data (at the bottom of the pyramid) provides the "what" and "why" of tasks, simulation robot data (at the middle) offers the "how" for robot control, and real-world robot data (at the top) bridges the "reality gap" between simulation and physical environments [39]. Their synergistic integration is therefore not merely beneficial but essential for pre-training robust, versatile, and human-aligned robot foundation models that can generalize across tasks, platforms, and physical environments. Accordingly, as illustrated in Table 1, we pre-train the model on a mixture of heterogeneous data sources:

- **Simulation data:** We collect simulated synthetic data are from InternData-A1 [31] and RoboTwin2.0 [10], providing diverse and low-cost manipulation trajectories for learning scene dynamics, object interaction, and coordinated control.
- **Human data:** We include egocentric human manipulation videos from EgoDex [13], which provide rich hand-object interaction priors and further enhance the model’s understanding of manipulation behaviors and scene evolution.
- **Real-world data:** The real-world robot data are drawn from RoboChallenge [35] and AgiBot-World [8], which help reduce the sim-to-real gap and improve robustness in physical environments.

The mixture of data sources covers 8 different robot embodiments and 300+ robot control tasks, including 50+ atomic manipulation skills such as grasping and placing. Notably, although the dataset provides sufficient diversity, it is not particularly large in scale: the total duration of all manipulation videos is only over 6,000 hours.

4 Experiments

In this section, we introduce comprehensive evaluation experiments, which are conducted to explore the following questions: 1) How does proposed TBot-SA₁ perform compared to existing VLA/WAM

Table 2: Real-robot experimental setup and task details. Trajectory Num./Time denotes the number of collected demonstration trajectories and the total data collection time in minutes.

Initial	Finished	Robotic Embodiment	Trajectory Num. / Time	Task Description
		Unimanual: AgileX PiPER (7-DoF)	30 / 21 min	RGB Block Sorting: Position red block, green block, and blue block from left to right in the specified sequence.
		Unimanual: AgileX PiPER (7-DoF)	50 / 39 min	Object Organization: Tidy up the table by sorting the toys, stationery, and trash into their designated containers.
		Bimanual: ARX Lift2 (14-DoF)	30 / 5 min	Trash Cleaning: Pick up the trash with the nearest gripper, hand it over to the other gripper, and place it into the trash bin.
		Bimanual: ARX Lift2 (14-DoF)	30 / 4 min	Pen Holder Placement: Pick up the stationery on the table and place it into the pen holder.
		Bimanual: ARX Lift2 (14-DoF)	31 / 4 min	Toy Box Organization: Pick up the toys from the desktop, place them into the toy box, and arrange them neatly inside.
		Bimanual: ARX Lift2 (14-DoF)	89 / 31 min	Sweep Trash: Sweep the trash into the dustpan using a broom.
		Bimanual: ARX Lift2 (14-DoF)	30 / 8 min	Unzip Pencil Bag: Grasp the pencil bag, unzip it, and place it back onto the tabletop.

models across multiple benchmarks and embodiments? 2) How does the design choice in proposed WSA paradigm contribute to robot learning performance? Next, we first present experimental setup. Then, we demonstrate the merit of TBot-SA₁ through a comparison with state-of-the-arts on both simulation and real-world tasks. Finally, we conduct ablation studies to illustrate major properties of proposed WSA paradigm.

4.1 Experimental Setup

Real-robot setup Real robot experiments are conducted with two embodiments: (1) Unimanual AgileX PiPER with three cameras and 6-DoF robotic arm; (2) Bimanual ARX Lift2 with three cameras and two 7-DoF robotic arms. To ensure diversity of robot behaviors in real-world scenario and investigate cross-embodiment generalization, we design 7 tabletop manipulation tasks. For each task, objects are randomly initialized within the workspace to evaluate model’s generalization capability across object position, orientation, and scene layout. The robot receives language instructions and visual observations as input, and predicts continuous action chunks for closed-loop execution. For testing, we adopt two metrics for comprehensive evaluation of RFMs: (1) Success Rate (SR), the proportion of rollouts in which the task is successfully completed. (2) Completeness (C), the proportion of rollouts in which the sub-task is completed.

Simulation setup We adopt RoboTwin-2.0 benchmark [10] for evaluation in simulation environment. RoboTwin platform is setup with easy and hard modes, where the former offers clean environment without any distracted factor and the latter conduct domain randomization with background changes and distracted objects. Following official protocol [10], 27500 demonstrations (2500 for clean, 25000 for hard) are adopted for model fine-tuning. In addition, we adopt multi-task test

Table 3: Real-world evaluation results on seven tabletop manipulation tasks. **SR** denotes the task success rate (%), and **C** denotes the task progress score (%).

Method		π_0		$\pi_{0.5}$		InternVLA-A1		TBot-SA ₁	
Model Type & Model Size		VLA	3B	VLA	3B	WAM	3B	WSA	3B
<i>Evaluation Results on 7 Real-world Tabletop Robot Manipulation Tasks</i>									
Task Name	Robot Type	SR	C	SR	C	SR	C	SR	C
RGB Block Sorting	AgileX Piper	70	70	70	70	50	50	80	80
Object Organization	AgileX Piper	50	50	77	77	37	37	80	80
Trash Cleaning	ARX Lift2	37	53	77	87	43	58	80	90
Pen Holder Placement	ARX Lift2	7	28	70	77	33	38	87	90
Toy Box Organization	ARX Lift2	13	18	13	27	37	48	60	72
Sweep Trash	ARX Lift2	30	40	27	44	44	78	80	89
Unzip Pencil Bag	ARX Lift2	30	41	55	63	25	43	75	75
Average (%)		34	36	55	59	38	53	77	83

setting (i.e., train for once, test for all tasks) to evaluate how much versatility of each model attain. During testing, success rate is adopted as the evaluation metric following the official protocol.

4.2 Main Results on Real-World Tasks

Real-world Evaluation Tasks As illustrated in Table 2, the real-world benchmark consists of 7 representative tabletop scenarios: RGB Block Sorting, Object Organization, Trash Cleaning, Pen Holder Placement, Toy Box Organization, Sweep Trash and Unzip Pencil Bag. These tasks require the robot to perceive cluttered scenes, interact with objects of different categories and spatial layouts, and complete multi-step manipulation behaviors such as grasping, placing, pushing, and sweeping. Compared with simple pick-and-place tasks, these scenarios place higher demands on spatial reasoning, long-horizon execution, and robustness to diverse object arrangements.

Baselines We compare TBot-SA₁ with three representative and open-source embodied foundation models: π_0 , a generalist policy for continuous action prediction; $\pi_{0.5}$, a policy for open-world robot control that demonstrates impressive generalization; and InternVLA-A1, a unified model integrating visual understanding, video generation and action generation. For evaluation, we report the mean task success rate and the task progress score. The success rate measures whether the whole task is completed successfully, while the progress score reveals progress of full task execution.

Evaluation Results As shown in Table 3, TBot-SA₁ achieves leading performance over baselines under real-world setting. Across 7 tabletop manipulation tasks, TBot-SA₁ achieves an average success rate of 77% and an average completeness score of 83%, substantially outperforming the second-best model $\pi_{0.5}$, which obtains 55% and 59%, respectively. Moreover, the improvement is consistent across both the single-arm PiPER and bimanual Lift2 platforms, with TBot-SA₁ achieving the highest success rate on each task. These results highlight the effectiveness of the proposed WSA architecture. Rather than relying only on reactive 2D visual-action mapping, TBot-SA₁ couples 3D-aligned visual thinking, 3D world dynamics, and 3D-conditioned action generation, leading to more reliable execution in spatially complex and long-horizon tasks.

4.3 Main Results on Simulation Benchmark

Simulation Tasks RoboTwin2.0 [24, 10] is a challenging dual-arm manipulation benchmark that evaluates policy generalization under both clean and randomized environments. It covers diverse bimanual skills, including object handover, coordinated manipulation, spatially constrained interaction, and long-horizon tabletop tasks.

Evaluation Results As shown in Table 4, TBot-SA₁ achieves the best performance across open-source models, obtaining a success rate of 93% in the hard setting. Compared with large WAM

Table 4: RoboTwin2.0 benchmark experimental results. TBot-SA₁ is evaluated against closed-source and open-source embodied foundation models under hard setting. Mean success rate **SR** over 50 manipulation tasks is reported. TBot-SA₁ achieves leading performance over open-sourced models.

Model Name	Access Status	Model Size	Model Type	SR (%)
Qwen-VLA [32]	Closed-source	5B	VLA	87
Being-H0.7 [23]	Closed-source	3B	WAM	90
MotuBrain [28]	Closed-source	Unknown	WAM	96
π_0 [7]	Open-source	3B	VLA	58
$\pi_{0.5}$ [6]	Open-source	3B	VLA	77
ABot-M0 [37]	Open-source	4B	VLA	85
Motus [4]	Open-source	8B	WAM	87
InternVLA-A1 [9]	Open-source	3B	WAM	88
LingBot-VA [17]	Open-source	7B	WAM	92
Fast-WAM [41]	Open-source	6B	WAM	92
TBot-SA₁(Ours)	Open-source	3B	WSA	93

Table 5: Comparison with state-of-the-art open-source models on RoboTwin2.0 under clean and randomized settings. Per-task success rate is reported.

Method	$\pi_{0.5}$ [6]		InternVLA-A1 [9]		Motus [4]		FastWAM [41]		TBot-SA ₁	
Type & Size	VLA	3B	WAM	3B	WAM	8B	WAM	6B	WSA	3B
<i>Evaluation Results on 50 Simulated Bimanual Robot Manipulation Tasks</i>										
Task Name	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
<i>Place Dual Shoes</i>	75	75	91	88	93	87	94	88	95	94
<i>Move Stapler Pad</i>	56	42	52	67	83	85	77	64	65	75
<i>Stack Blocks Two</i>	97	100	100	100	100	98	100	100	100	100
<i>Scan Object</i>	72	65	86	83	67	66	89	92	85	88
<i>Place Object Stand</i>	91	85	89	88	98	97	90	94	95	95
<i>Place Fan</i>	87	85	98	98	91	87	96	96	98	99
<i>Move Pillbottle Pad</i>	84	61	98	99	93	96	100	99	99	100
<i>Pick Dual Bottles</i>	93	63	92	87	96	90	100	96	94	89
<i>Blocks Ranking Rgb</i>	92	85	92	92	99	97	100	100	95	95
..... (50 tasks)										
<i>Turn Switch</i>	62	54	56	62	84	78	61	59	61	75
<i>Pick Diverse Bottles</i>	81	71	77	77	90	91	80	85	86	76
<i>Place Bread Basket</i>	77	64	90	91	91	94	91	93	99	94
<i>Stack Blocks Three</i>	91	76	90	93	91	95	95	97	98	98
<i>Put Bottles Dustbin</i>	84	79	90	93	81	79	95	90	95	92
<i>Place Can Basket</i>	62	62	81	72	81	76	71	69	80	79
<i>Stamp Seal</i>	79	55	75	75	93	92	90	94	73	82
<i>Hanging Mug</i>	18	17	31	47	38	38	58	62	50	50
<i>Handover Block</i>	66	57	90	73	86	73	95	81	91	84
<i>Stack Bowls Three</i>	77	71	89	90	79	87	80	81	92	93
<i>Place Object Basket</i>	80	76	89	87	81	87	89	88	84	87
<i>Open Microwave</i>	34	77	100	99	95	91	62	45	100	100
<i>Average (%)</i>	83	77	89	88	89	87	92	92	92	93

models such as *Motus* and *Lingbot-VA* (model size $> 3B$), TBot-SA₁ with a model size of 3B delivers higher performance, and meanwhile remains competitive with the best closed-source model *MotuBrain*. As illustrated in Table 5, TBot-SA₁ exhibits generalizable capability on long-horizon and spatially demanding tasks such as *Move Pillbottle Pad*, *Stack Blocks Three*, and *Open Microwave*. These results suggest that the proposed WSA paradigm is beneficial for learning transferable 3D world-action priors that support robust bimanual manipulation under diverse conditions.

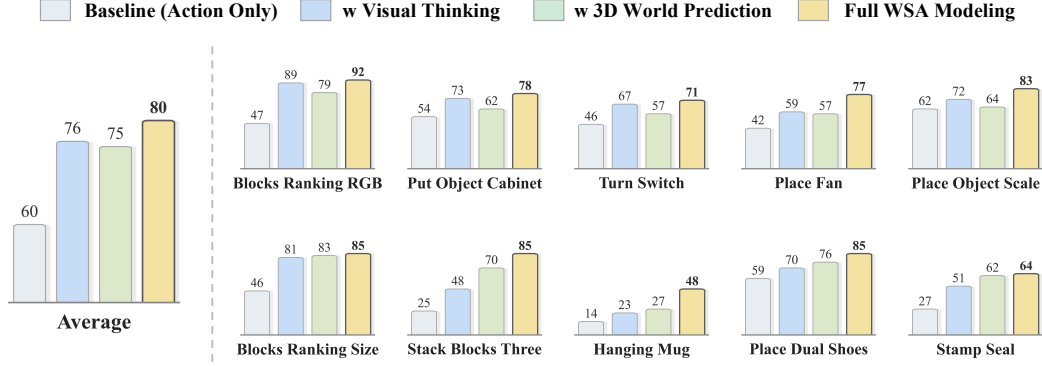


Figure 4: Investigation of WSA modeling on RoboTwin2.0 Benchmark.

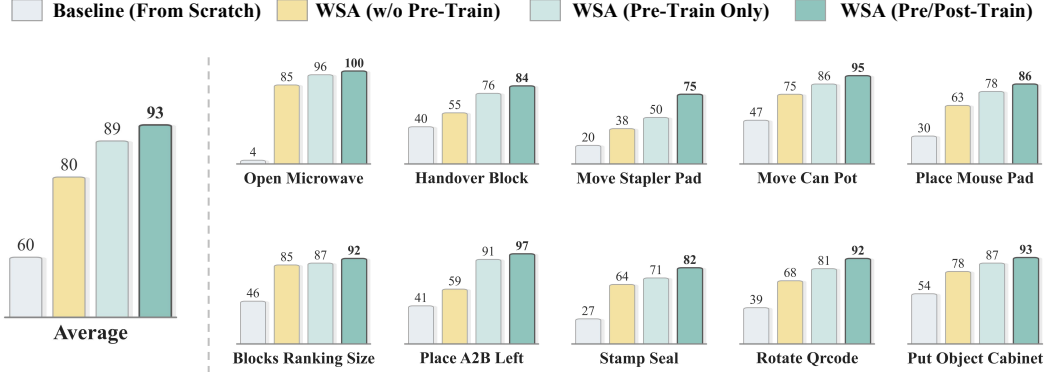


Figure 5: Ablation studies of WSA pre-training on RoboTwin2.0 Benchmark.

4.4 Ablation Study

In this section, we seek to answer the second question via two ablative experiments. First, we perform ablation studies to identify the contribution of causal modeling in WSA, which involves three learning tasks: visual planning, 3D world prediction and 3D action generation. Second, we study the effect of pre-training to reveal the details of full training process in WSA.

Investigation of WSA modeling We adopt the π -style VLA policy proposed in prior works [7, 6] as the baseline model, which integrates the pre-trained QWen3-VL 3B vision-language model [2] with a randomly initialized diffusion transformer serving as the action expert. All ablation experiments are conducted on the RoboTwin2.0 platform, and the corresponding results are presented in Figure 4. Specifically, the baseline model achieves a success rate (SR) of 60%. Notably, learning visual planning improves the SR of action generation to 76%, while learning 3D world model yields a competitive SR of 75%. Full WSA modeling framework further outperforms all individual variants, reaching the best SR of 80%. Besides, modeling visual planning is beneficial for manipulation tasks that rely heavily on visual semantics, object appearance, and temporally observable state changes, such as picking bottles and open microwave. In contrast, learning 3D world model improves contact-rich manipulation tasks such as rotate QR Code and stack blocks, which require a deep understanding of spatial geometry, object pose, and hand-object interaction. Based on the above observations, two key conclusions can be drawn. First, learning world dynamics in both 2D and 3D latent space is essential for generating generalized robot actions. Second, the bidirectional 3D world-action causal mechanism imposes effective constraints on action generation, thereby significantly enhancing the robustness of robot action learning.

The effect of pre-training To comprehensively investigate the efficacy of WSA pre-training, we design three comparative ablation settings: (1) WSA modeling is only applied in the post-training stage without pre-training on diverse heterogeneous robot datasets; (2) WSA modeling is adopted in

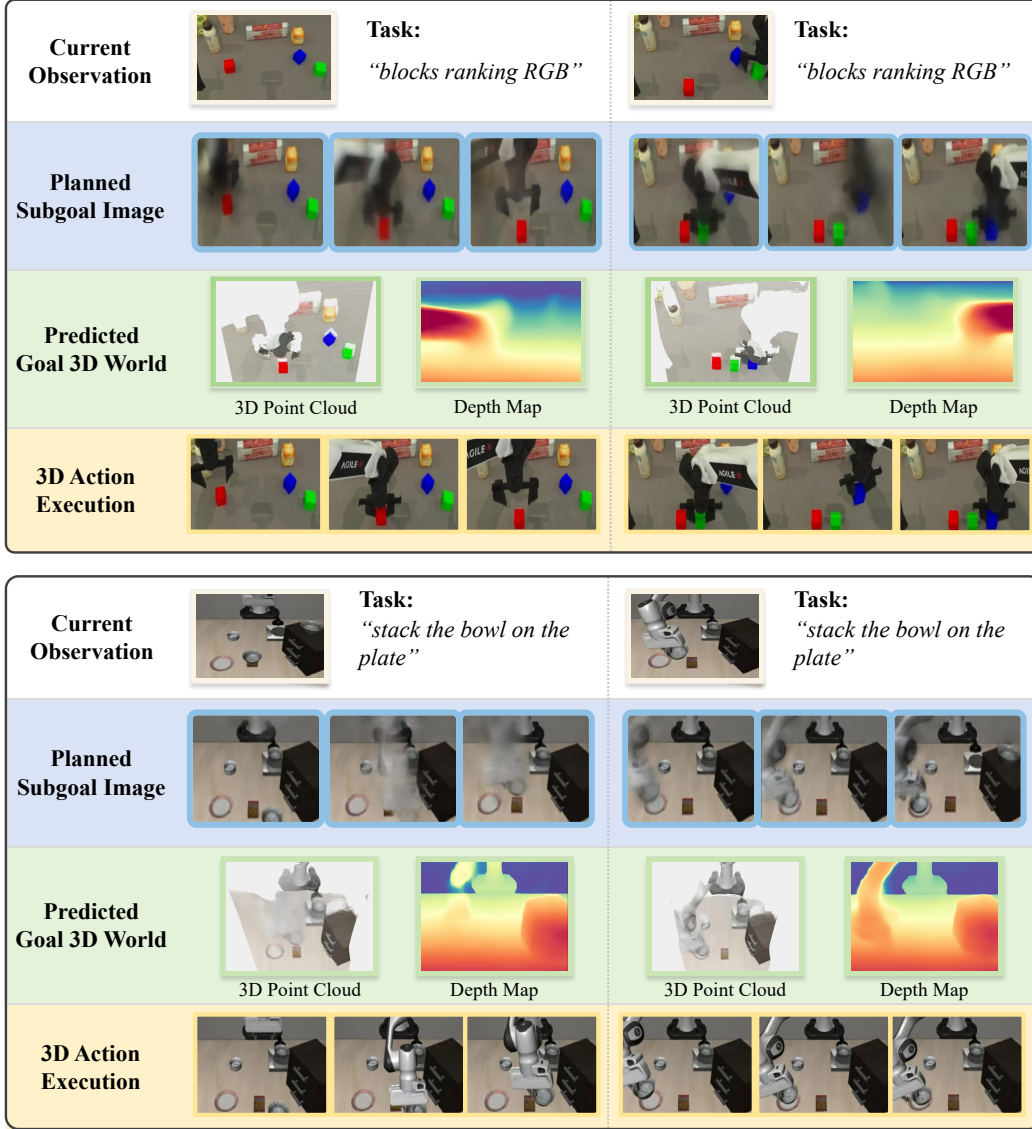


Figure 6: Visualization examples generated from TBot-SA₁. First row: the current observations with text instruction. Second row: generated subgoal images for visual planning. Third row: predicted 3D world that is rendered via Depth Anything model [19]. Fourth row: the real trajectory of executed actions.

the pre-training stage, while the post-training process only optimizes the action generation objective; (3) the model is consistently optimized with WSA modeling in both pre-training and post-training stages. As summarized in Figure 5, WSA pre-training delivers a substantial performance gain, increasing the SR from 80% to 89%. Moreover, applying WSA modeling in both pre-training and post-training further boosts the SR to 93%. These results demonstrate that WSA serves as an effective and essential training paradigm for building robust robot foundation models.

4.5 Qualitative Analysis

To reveal how TBot-SA₁ performs robot control, we present qualitative visualizations of the model’s predictions on various tasks. As illustrated in Figure 6, the visualization covers four views: visual observation with text instruction, predicted visual thought, goal 3D world prediction with 3D Gaussian and depth map, and trajectory of executed actions. From the results, we find that predicted visual

plans well capture the task objective and well align the generated 3D world. Furthermore, driven by explicit 3D world modeling, the action trajectories are well-aligned with the predicted 3D goal states. These visual results demonstrate that TBot-SA₁ unifies semantic visual planning, causal 3D world dynamics, and closed-loop action generation within a shared latent space. It explicitly infers how actions alter the 3D world and how 3D world states guide action generation, thus leading to robust and generalizable manipulation behaviors.

5 Conclusion

This work addresses critical limitations of current robot foundation models paradigms, which lack understanding of 3D physical dynamics and bidirectional world–action causality. We propose the 3D-Causal World-Spatial-Action (WSA) paradigm and introduce TBot-SA₁, a novel robot foundation model for generalizable robot control. Built on Mixture-of-Transformers architecture and 3D-centric bidirectional causal mechanism, proposed TBot-SA₁ jointly learns 3D world-aware visual thinking, action-caused 3D world modeling, and 3D inverse dynamics modeling in a unified latent space. The model is pre-trained with only 6,000 hours of demonstration data (1,000 hours from real robots), yet achieves strong performance across simulation and real-world benchmarks. Our work demonstrates that modeling 3D world–action bidirectional causality enables robots to inherently acquire physical commonsense and causal mechanisms, paving a practical and affordable path for generalizable robot foundation model.

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